

Efe Şakrak

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Kusadası / Aydın

linkedin.com/in/sakrakefe 12/07/2004

SUMMARY

A Computer Engineer dedicated to writing clean, maintainable code with a strong focus on user experience. I aim to continuously update my skills in a fast-paced tech environment and become a valuable asset to innovative teams.

LANGUAGES

Turkish ●●●●●

English ●●●●○

SKILLS

- **Frontend Technologies:** React.js, Vite, Tailwind CSS, JavaScript (ES6+), HTML5, CSS3
- **Architecture & Integration:** Headless CMS, Dynamic Content Management, Custom Admin Dashboards, RESTful APIs
- **Engineering Core & Languages:** C++, Java, Python Object-Oriented Programming (OOP), Data Structures & Algorithms
- **Tools & Version Control:** Git, GitHub, VS Code
- **Design & User Experience:** Responsive / Mobile-First Design, UI/UX Principles, [Figma]

INTERESTS

Active Musician

- Vocalist & Guitarist in an alternative rock band, Live Performance
- Production (DAW), Conceptual Storytelling, Digital Audio Technology, Modern Web Aesthetics (UI/UX).

CERTIFICATES

- Advanced Python Programming Certificate

EDUCATION

Graduate
National Technical University Kharkiv Polytechnic Institute
2022 – 2026 | Ukraine
Computer Engineering

PROJECTS

SolarTrack: AI-Driven IIoT Solar Power Monitoring Platform
Full-Stack Developer Tech Stack: React.js, FastAPI, Python, Scikit-Learn, SQLite, SQLAlchemy, Tailwind CSS

- **Architected** a decoupled, N-Tier Industrial IoT (IIoT) monitoring platform to transition solar facility management from reactive to predictive maintenance.
- **Developed** an asynchronous backend REST API using FastAPI, capable of handling high-frequency sensor telemetry with an ultra-low latency of 18ms and 0% data loss under 500 RPS peak load.
- **Built** a high-performance, real-time frontend dashboard utilizing React.js and Recharts, leveraging Virtual DOM optimizations to render continuous data streams at a steady 60 FPS without memory leaks.
- **Integrated** a Scikit-Learn Linear Regression engine directly into the edge-computing backend to generate 15-step short-term power forecasts, achieving an R^2 accuracy score of 0.89.
- **Engineered** a Python-based stochastic emulator using Gaussian noise algorithms to mathematically simulate physical hardware fluctuations of a 3.5 kW solar array.
- **Secured** the data pipeline against SQL injection and malformed payloads using SQLAlchemy ORM parameterization and Pydantic schema validation.

AI-Powered Text-to-Image Generator
Developed an application that integrates with an AI API to process user text prompts and simultaneously generate a grid of 4 unique, high-quality image variations.

Full-Stack Web Project (Pre-Deployment)
Designed and developed a comprehensive, responsive website applying modern web technologies (React, Tailwind), currently fully functional and prepped for production deployment.

Low-Level System Controller: BIOS LED Modification
Engineered a system-level script interfacing with the system BIOS to override standard keyboard behavior, repurposing the Caps Lock LED into a customizable blinking indicator.